

A monotone hierarchy of upper bounds from one flowed correlator, and why no bound of this kind can give $m > 0$

Let φ_t be any flowed gauge-invariant local scalar with $\langle \varphi_t \rangle = 0$ and let

$$C_0(\tau) = \int d^3z \langle \varphi_t(z, \tau) \varphi_t(0, 0) \rangle^c = \int_{[m, \infty)} d\rho(E) e^{-E\tau/\hbar}, \quad \rho \geq 0,$$

be its zero-spatial-momentum Euclidean correlator, with ρ the Källén–Lehmann measure supplied by Osterwalder–Schrader reconstruction. Writing $M_k = \int E^k d\rho(E)$, every consecutive ratio is an upper bound for the mass gap and the bounds improve monotonically as the index decreases:

$$m \leq \dots \leq \frac{M_{-1}}{M_{-2}} \leq \frac{M_0}{M_{-1}} \leq \frac{M_1}{M_0}, \quad \text{with} \quad \frac{M_1}{M_0} \text{ the } f\text{-sum bound of}$$

the finiteness note, which is therefore the weakest member of the family. The first member below it is

$$m \leq \frac{M_0}{M_{-1}} = \frac{\hbar C_0(0)}{\int_0^\infty C_0(\tau) d\tau}$$

and it needs strictly less than the f -sum bound: no gradient term, no equal-time Hamiltonian smearing, hence no hypothesis (F1). Both quantities are integrals of the four-dimensional flowed correlator that Lüscher’s renormalization statement covers, so the only inputs are the existence of the theory with the Osterwalder–Schrader axioms and $C_0 \not\equiv 0$. In free Maxwell theory the first three members take the values 1.504, 1.329, 1.128 in units of $\hbar c/\sqrt{t}$, decreasing as the hierarchy requires. The negative moments diverge when the spectral measure reaches down to $E = 0$, so their finiteness is necessary for a gap; it is not sufficient, since $d\rho = e^{-\hbar c\kappa/E} dE$ has every negative moment finite with $\inf \text{supp } \rho = 0$. Consequently **no member of this hierarchy, and no upper bound of this type, can establish $m > 0$** : the entire upper side of the problem settles the finiteness clause and is silent on the clause that matters. Constants explicit; nothing promoted.

1. Moments of the flowed correlator

Assume the continuum theory exists with the Osterwalder–Schrader axioms, a unique vacuum and a self-adjoint $H \geq 0$ with $H\Omega = 0$; let φ_t be a gauge-invariant local scalar in the field at flow time $t > 0$, with $\langle \varphi_t \rangle_0 = 0$. Reconstruction gives, for the zero-spatial-momentum correlator at Euclidean separation $\tau > 0$,

$$C_0(\tau) = \int d^3z \langle \varphi_t(z, \tau) \varphi_t(0, 0) \rangle_0^c = \langle \Omega, \Phi^\dagger e^{-\tau H/\hbar} \Phi \Omega \rangle = \int_{[m, \infty)} e^{-E\tau/\hbar} d\rho(E),$$

where Φ is the zero-momentum smeared operator of the finiteness note §2 and $d\rho$ is the spectral measure of H in the state $\Phi\Omega$, divided by the volume so that ρ is intensive. The vacuum contribution is removed by the connected part, so $\text{supp } \rho \subset [m, \infty)$ where $m = \inf(\text{spec } H \setminus \{0\})$. Set

$$M_k = \int_{[m, \infty)} E^k d\rho(E) \in [0, \infty], \quad k \in \mathbb{Z}.$$

Two of them have elementary readings:

$$M_0 = C_0(0^+), \quad M_{-1} = \frac{1}{\hbar} \int_0^\infty C_0(\tau) d\tau, \quad M_1 = -\hbar \left. \frac{dC_0}{d\tau} \right|_{\tau=0^+},$$

the equal-time zero-momentum variance, the time-integrated correlator, and the f -sum rule.

2. The hierarchy

Proposition 1. If M_k and M_{k+1} are finite and $M_k > 0$, then

$$m \leq \frac{M_{k+1}}{M_k}.$$

If in addition $M_{k-1} < \infty$ and $M_{k-1} > 0$, then

$$\frac{M_k}{M_{k-1}} \leq \frac{M_{k+1}}{M_k}.$$

Proof. On $\text{supp } \rho$ one has $E \geq m$, so $M_{k+1} = \int E \cdot E^k d\rho \geq m \int E^k d\rho = m M_k$, which is the first claim. For the second, Cauchy–Schwarz in $L^2(d\rho)$ applied to $E^{(k-1)/2}$ and $E^{(k+1)/2}$ gives $M_k^2 = (\int E^{(k-1)/2} E^{(k+1)/2} d\rho)^2 \leq M_{k-1} M_{k+1}$, that is, (M_k) is log-convex; dividing by $M_{k-1} M_k$ gives the claim. $\square$

The chain therefore runs downward from the f -sum bound:

$$m \leq \dots \leq \frac{M_{-1}}{M_{-2}} \leq \frac{M_0}{M_{-1}} \leq \frac{M_1}{M_0}.$$

Each step uses one more inverse power of the energy and so weights the low-lying states more heavily; in the limit of small index the ratio converges to $\inf \text{supp } \rho$ whenever the moments stay finite, by dominated convergence applied to ρ restricted to $[m, m + \epsilon]$ against its complement.

3. The first improvement, and what it costs

Corollary 2. If $0 < M_{-1}$ and $M_0 < \infty$, then

$$m \leq \frac{M_0}{M_{-1}} = \frac{\hbar C_0(0^+)}{\int_0^\infty C_0(\tau) d\tau} < \infty.$$

Compare the inputs.

bound	needs	supplied by
M_1/M_0	$\langle \delta O / \delta A ^2 \rangle < \infty$, equal-time spatial smearing	hypothesis (F1) of the finiteness note
M_0/M_{-1}	$C_0(0^+) < \infty$ and $\int_0^\infty C_0 > 0$	four-dimensional flow: Lüscher’s statement

The f -sum bound required the Hamiltonian counterpart of the flow’s renormalization, because M_1 involves the functional derivative of the operator with respect to the equal-time field. The ratio M_0/M_{-1} involves only the flowed correlator itself, and Lüscher states that the gauge field at flow time $t > 0$ is a smooth renormalized field, so that expectation values of local gauge-invariant expressions in it are well-defined physical quantities (arXiv:1006.4518v3, abstract and §2; passage level via the local companion). Hypothesis (F1) is therefore not needed for Corollary 2, and the remaining inputs are the existence of the theory with the axioms, which is T4 of the obligations map, and $C_0 \neq 0$.

Proposition 3 (the finiteness half, restated). Assume T4 and that some flowed local gauge-invariant scalar has $C_0 \neq 0$ with $C_0(0^+) < \infty$. Then $m < \infty$.

Proof. $C_0 \neq 0$ and $\rho \geq 0$ give $M_{-1} > 0$; Corollary 2 applies. $\square$

The clause $m < \infty$ of the Jaffe–Witten statement is thus equivalent, given the construction, to the nontriviality of a single flowed correlator, which is the sharpest form this programme has reached.

4. Free-field check of the monotonicity

For free Maxwell theory with φ_t the flowed magnetic energy density, the free-field note gives the spectral weight $d\rho \propto k^4 e^{-4tk^2} dk$ at energy $E = 2\hbar ck$. With $\int_0^\infty k^n e^{-4tk^2} dk$ evaluated by $\int_0^\infty k^{2j} e^{-\alpha k^2} dk = \frac{(2j-1)!!}{2(2\alpha)^j} \sqrt{\pi/\alpha}$ and $\int_0^\infty k^{2j+1} e^{-\alpha k^2} dk = \frac{j!}{2\alpha^{j+1}}$ at $\alpha = 4t$:

$$\begin{aligned}\frac{M_1}{M_0} &= 2\hbar c \frac{\int k^5 e^{-4tk^2}}{\int k^4 e^{-4tk^2}} = \frac{8}{3\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}} \approx 1.504 \frac{\hbar c}{\sqrt{t}}, \\ \frac{M_0}{M_{-1}} &= 2\hbar c \frac{\int k^4 e^{-4tk^2}}{\int k^3 e^{-4tk^2}} = \frac{3\sqrt{\pi}}{4} \frac{\hbar c}{\sqrt{t}} \approx 1.329 \frac{\hbar c}{\sqrt{t}}, \\ \frac{M_{-1}}{M_{-2}} &= 2\hbar c \frac{\int k^3 e^{-4tk^2}}{\int k^2 e^{-4tk^2}} = \frac{2}{\sqrt{\pi}} \frac{\hbar c}{\sqrt{t}} \approx 1.128 \frac{\hbar c}{\sqrt{t}},\end{aligned}$$

using $\int k^5 = 1/(64t^3)$, $\int k^4 = 3\sqrt{\pi}/(256t^{5/2})$, $\int k^3 = 1/(32t^2)$, $\int k^2 = \sqrt{\pi}/(32t^{3/2})$. The three numbers decrease, as Proposition 1 requires, and the first reproduces the free-field note. In this theory M_k is finite for $k \geq -4$ and diverges for $k \leq -5$, since $d\rho \propto E^4 dE$ near $E = 0$: the hierarchy terminates, and its terminating member does not reach $m = 0$. Termination at a finite index is the spectral signature of weight accumulating at zero energy.

5. Why the upper side cannot give $m > 0$

Every bound above is of the form $m \leq (\text{ratio of moments})$, and each is finite whenever the corresponding moments are. Finiteness of all negative moments is necessary for $m > 0$ but not sufficient: take

$$d\rho(E) = e^{-\hbar c\kappa/E} dE \quad \text{near } E = 0, \quad \kappa > 0,$$

whose support reaches $E = 0$, so $m = 0$, while $M_{-k} = \int E^{-k} e^{-\hbar c\kappa/E} dE = (\hbar c\kappa)^{1-k} \Gamma(k-1)$ after $u = \hbar c\kappa/E$, finite for every $k \geq 2$. A theory with this spectral density would satisfy every bound in the hierarchy with a positive right-hand side and still be gapless. No amount of information about the moments of one correlator distinguishes it from a gapped theory.

This is not a defect of the flow or of the trial operator. The variational principle produces upper bounds, and $m > 0$ is a lower bound; the two require different mathematics, and the entire content of Sections 1–4, like that of the upper-bound note and the Polyakov note, belongs to the finiteness clause. The clause $m > 0$ needs an argument of the kind attempted on the lower side: an operator inequality $H \geq m(1 - |\Omega\rangle\langle\Omega|)$, which is what the strong-coupling theorem of the T2 note achieves at fixed cutoff and what the Schur note shows is obstructed by ultraviolet dressing in the continuum.

6. Consequence for STATE

The upper side is closed as far as it can go: a monotone hierarchy whose best members need only T4 and the nontriviality of one flowed correlator, with the f -sum bound identified as its weakest member and hypothesis (F1) eliminated. The finiteness clause is therefore reduced to the construction plus one nonvanishing correlator, and no further work on upper bounds changes the standing of the problem. Everything remaining is the lower bound: T2' at fixed cutoff and its uniformity, which is where the next steps belong.